Benchmarking Proposal

Company XYZ Data Centres

A More Efficient and Effective Benchmarking Initiative
as Part of a TQM Strategy

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The Starting Point – What the Gartner Case Tells Us

Our mandate (Gartner Bottom Line): "...such projects are not universally desirable" — as external consultants, we propose a better way forward.

3 Regions

EMEA, APAC, Americas — all run simultaneously

3 Months

Compressed timeline — too little time to analyse properly

Top 10%

Results achieved — but at too high a cost and risk

What XYZ's Experience Revealed:

› Simultaneous studies were extremely stressful — too much happening at once

› Little time to analyse trends and differences between data centres

› Urgency introduced risk — things could have gone wrong more easily

› Data collection quality suffered across regions without a controlled approach

"If we were to do it again, we would stagger the regional projects." **— Company XYZ Project Director**

Key Issues Identified – Why Simultaneous Studies Failed

01

Resource Overstretch

Running 3 regions in parallel drained consultant bandwidth and internal management capacity simultaneously

02

No Opportunities to Learn Between Studies

Staggering allows lessons from Region 1 to improve data collection design in Region 2 and 3

03

Inconsistent Data Quality

Simultaneous studies produce inconsistent data collection standards across regions without a controlled approach

04

Change Barriers Not Addressed

Rapid simultaneous implementation left insufficient time to manage resistance and embed change per region

Our Proposed Approach – PDCA Embedded in TQM Strategy

PLAN

Select subject: Data Centre PUE (Power Usage Effectiveness)
Define milestones
FMEA pre-screening
Choose benchmarking type
Set measures

DO

Conduct benchmark visits

Pre-Visit Preparatory Sheet

Code of Conduct

Collect data · MoM

CHECK

Analyse gap (How big? How applicable?)

Performance metrics vs practice
Top 10% RPN threshold

ACT

Implement change
Lewin 3-stage
House of Change

Improvement is an ongoing process · Benchmarking becomes part of TQM strategy, not a one-off exercise (Bendell, 1997)

Selecting the Subject – The Four Questions

Where are we now?

Company XYZ data centres performing in top 10% of Gartner database — but simultaneous study approach was unsustainable and high risk.

Where do we want to go?

Consistent, world-class PUE (Power Usage Effectiveness) across all regions with repeatable process

What do we need to do to get there?

Staggered Functional Benchmarking study, one region per year, applying PDCA and FMEA screening

Did we get there?

Annual review via PDCA Check phase — calibrate benchmark continuously (Xerox model)

Chosen Type

Functional Benchmarking

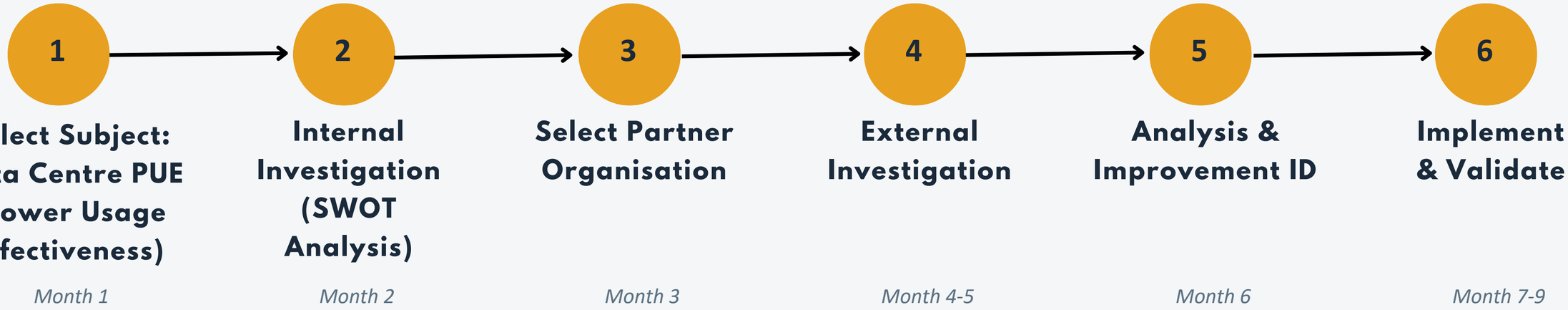
Compare with non-competitors carrying out same functions across industries

Why not Competitive?
Data sensitivity in DC industry makes competitor data difficult to obtain ethically (Code of Conduct §6.2)

Year 1 Focus:
Power Usage Effectiveness (PUE)

Planning the Project – Milestones and Start-Up

Staggered: Year 1 – EMEA | Year 2 – APAC | Year 3 – Americas



Project Start-Up Checklist (Bendell)

✓ Clear project definition and purpose?

✓ Measures relate to project, process and purpose?

✓ Critical path identified?

✓ Clear process definition?

✓ Are measures measurable?

✓ Fallback plan in place?

Data Collection – Process Mapping and FMEA

Illustrative FMEA - Data Center Benchmarking Process

Failure Modes and Effects Analysis (FMEA)

Pre-screen the benchmarking process to identify highest-risk failure points before the visit.

RPN = Severity (S) × Occurrence (O) × Detection (D) **Max = 1,000**

Failure Mode	S	O	D	RPN
Wrong benchmarking method selected	9	7	5	315 ⚠
Wrong briefing of partner organisation	9	7	5	315 ⚠
Poor quality of partner's first interview	9	5	7	315 ⚠
Poor quality of applications/data	9	5	5	225 ⚠
Incorrect data input / system update	9	5	5	225 ⚠

Threshold: Top 10% RPN = 225 → Priority action required

Pareto Principle

Focus remediation on the top 20% of failure modes that drive 80% of risk. RPN ≥ 225 = mandatory action before visit proceeds.

Process Mapping

Map the internal data centre benchmarking process before any external contact is made.
This enables asking the right questions during the visit (Bendell, 1997).

S = 1 (none) → 10 (hazard)
O = 1 (never) → 10 (almost certain)
D = 1 (certain) → 10 (impossible to detect)

Benchmark Visit Preparation – Pre-Work and The Team

Pre-Work

› Be clear about the WHY before starting any visit

› Thoroughly investigate your own process first

› Desk research is essential — annual reports, databases, internet

› Initial phone call: inform partner of goals, explain your process

› Send Pre-Visit Letter: Who are we / What we offer / What we want to know / Questionnaire

› Agree Pre-Visit-Preparatory-Sheet with partner

› Take legal advice before sending any questionnaires (Code of Conduct §1.4)

The Team

- **Process Owner**
  Leads the visit
- **Customer of the Process**
  End-user perspective
- **Supplier of the Process**
  Input perspective
- **MoM Coordinator**
  Minutes of the meeting
- **Question Owner**
  KISS – Keep It Short and Simple

Source: Bendell, Tony (1997)

Conducting the Visit – Code of Conduct and Protocol

During the Visit

- ✓ Stick to the agenda and questions previously agreed
- ✓ Behave according to the European Benchmarking Code of Conduct
- ✓ Stick to the time schedule
- ✓ Do not use company / industry jargon
- ✓ Do not discuss other companies or pricing with competitors
- ✓ Thank the host organisation
- ✓ Conduct debriefing on the same day as the visit
- ✓ Distribute and agree Minutes of Meeting (MoM)

European Code of Conduct
8 Principles

1. Preparation
2. Contact
3. Exchange
4. Confidentiality
5. Use
6. Legality
7. Completion
8. Understanding & Agreement

Source: European Benchmarking Code of Conduct (APQC/SPI / Performance Improvement Group)

Analysing the Data – Gap Analysis and Performance Metrics

Two Elements Must Be Analysed Together (Bendell, 1997):

WHAT is achieved

Performance Metrics (Numbers)

- PUE measurements by region
- Cost per kWh
- Uptime %
- Cooling efficiency ratios
- Carbon footprint per MW

HOW & WHY it is achieved

Practice and Process

- Cooling architecture used
- Staff training model
- Maintenance scheduling
- Procurement approach
- Change management in place

Two Key Questions (Bendell, 1997)

How big is the gap between their performance and ours?

How much of their experience is applicable to our situation?

Implementation and Change Management

Lewin's Three Stage Change Process

UNFREEZE

Disturb equilibrium · Challenge assumptions · Benchmark with outside organisations · Feedback on performance · Create urgency

MOVE

From old behaviours to new behaviours · Implement staggered approach · Train teams · Pilot changes on small scale first

REFREEZE

Establish new behaviour as normal · Stabilise by balancing driving and restraining forces · Communicate success · Keep measuring

House of Change

Contentment

No urgency — shock needed

Denial

Blame others — communicate big picture

Confusion

No direction — provide vision

Renewal

Clear objectives — flexible, creative

Change Barriers (7S / PEST)

Perceptual · Emotional · Cultural · Environmental · Cognitive · Competing initiatives

Communication Strategy – It Is Not Just E-mails

"It is not just e-mails..."

Communicate at All Levels

From executive sponsor to operational staff — no group excluded

Explain the Big Picture

Connect benchmarking initiative to TQM strategy and business goals

Show Respect for the Past

Acknowledge what worked before; position change as evolution not criticism

Look for Early Success Stories

Surface and amplify quick wins to build momentum and credibility

Celebrate Success

Recognise milestones publicly; reinforces Refreeze stage of Lewin model

Get Feedback

Two-way communication — listen as well as broadcast; adjust based on response

Source: PPT20 – Implementation / Communicating · Bendell, Tony (1997)

Benchmarking as Part of TQM Strategy

Embedding Benchmarking into TQM

Calibrating the Benchmark Continuously (Xerox; cited in Bendel - 1997)

Continuously revisit best practice — the benchmark itself moves. Do not treat any study as a one-off final answer.

Improvement is an Ongoing Process

Each PDCA cycle feeds the next. Year 1 EMEA findings improve Year 2 APAC study design.

EFQM Model Integration

Enablers (50%): Leadership, People, Strategy, Partnerships. Results (50%): Customer, People, Society, Business.

Marginal Management vs Benchmarking

Marginal: where are you vs last year? Benchmarking: where are you vs the market leaders?

EFQM Excellence Model

ENABLERS 50%

- Leadership
- People
- Strategy
- Partnerships & Resources
- Processes

RESULTS 50%

- Customer Results
- People Results
- Society Results
- Business Results

Conclusions and Recommendations

R1

Stagger Regional Studies

Year 1: EMEA · Year 2: APAC · Year 3: Americas. Apply lessons from each region to the next.

R2

Appoint Process Owner and Cross-Functional Team

Process owner + customer + supplier of process. Define roles, MoM coordinator, question owners (KISS).

R3

Run FMEA Pre-Screening Before Each Study

Identify highest RPN failure modes (≥225) and resolve before benchmark visit proceeds.

R4

Embed in TQM Strategy with Annual PDCA Review

Calibrate the benchmark continuously. Benchmarking is not a one-off — it is a living process.

R5

Strictly Follow European Code of Conduct

Especially when working with competitors: legal advice before contact, data blinded, no price sharing.

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